A. Finite-budget operation count

One approximate operation is a scalar Gaussian or finite uniform draw, one complete three-outcome IRS transition, or one weight coin. Its error includes the numerical evaluation needed for that draw and all of its bounded random draws. Guard bits are allocated within an operation so that its conditional total-variation error is at most 2^{-b} uniformly over the input. This is the premise of the finite-precision error bound $\delta_{\mathrm{samp}} \le R2^{-b}$ given in Eq. (5) in Section D of the supplementary proofs for Sections V and VI. A vector draw is counted coefficient by coefficient. A norm-conditioned Gaussian vector uses at most $b+3$ trials, as in the bounded sampler above; its cutoff comparison is included in the same allocation.

A deliberately loose common operation bound is

$$\begin{aligned} R_{\mathrm{count}} = 128\big[&4UL_k(Dn + mln + mn + d + 4) \\ &+ S((b+3)Dn + 2d + 8) \\ &+ Q(mln + mn + d + 4)\big]. \end{aligned}$$

The first term counts base and auxiliary secret trials, public-matrix draws, local uniform normal-form instances and initialization coins. The second counts every possible ring or auxiliary branch attempt, its bounded Gaussian-vector trials, at most d IRS transitions and the weight coin. The last term overcounts all oracle outputs coefficient by coefficient, including deterministic programmed entries and all UK bulk installations. All repeated and preprocessing queries were already included in Q.

The factor 128 provides common slack for the bounded sampler-bearing comparison games and their couplings. There are at most eight such versions in a displayed subreduction: real signing, capped independent challenges, private rejected entries erased, direct accepting simulation, norm-conditioned keys, unconditioned NTWE keys, uniform keys and the normal-form extraction game. Either candidate hybrid in anonymity and either primitive branch in unforgeability may each be charged separately; $8 \cdot 2 \cdot 2 < 128$. Replaying a fork is already included in the aggregate resource bounds, so it is not an omitted extra execution. The unbounded analytical continuation used only in the simulation coupling is not run by a computational reduction, and its cutoff event is charged to δ_{cap} instead. The count also covers the simpler bounded versions that skip rejected signing trials.

For the illustration in Section VII-B1 of the main paper, exact integer evaluation gives $R_{\mathrm{count}} < 2^{75} < R = 2^{88}$. For the wider query envelope considered below, the maximum count is still below 2^{88}. Thus the same value of R can be used in both comparisons.

1) Conditional numerical error propagation: For the finite-budget illustration in Section VII-B1 of the main paper, the operation count above with $R = 2^{88}$ gives $\delta_{\mathrm{samp}} \le 2^{-232}, 2^{-296}, 2^{-360}$, conditionally on the per-operation total-variation premise. The new-tag hash-guessing term $KQ^2/2^d$ is exactly $2^{-140}, 2^{-220}, 2^{-372}$. After propagating all comparison terms through Theorems 1–4, the sum of their four non-hardness parts is below $2^{-138}, 2^{-218}, 2^{-304}$, respectively. These figures include the tighter candidate norm-tail and invertibility bounds, the key-generation and signing cutoffs, oracle simulation, collisions and finite-precision comparisons.

For example, in this envelope the coefficient of $\sqrt{\epsilon_{\mathrm{M}}}$ in the new-tag bound is about $2^{121.03}$, and the NTWE coefficient in the anonymity bound is 2^{33}. To make each of these contributions at most a quarter of 2^{-t}, sufficient primitive bounds for $t = 120, 180, 256$ are $\epsilon_{\mathrm{N}} \le 2^{-155}, 2^{-215}, 2^{-291}$ and $\epsilon_{\mathrm{M}} \le 2^{-487}, 2^{-607}, 2^{-759}$, respectively, at the augmented reductions' running times. These required advantage bounds use the conditional interpretation in Section VII-B1 of the main paper.

The dependence on the envelope is material. With aggregate external caps $q_G, q_R, q_A \le 2^{48}$, the same bulk accounting is valid with $K = 2^{50}$ and $Q = 2^{67}$. The new-tag hash terms then become $2^{-96}, 2^{-176}, 2^{-328}$. Consequently the first two profiles do not meet their label targets even for the non-hardness part of this more permissive illustration. These are upper-bound limitations, not attacks. The accompanying executable budget check records both envelopes and every affected error and reduction coefficient.